Algo 8 Reverse Engineering Complete Research Report

Structure, exit engine, allocation reconstruction, machine learning, OHLCV foundation and alpha status

Prepared for: Rahul (Quant Researcher)

Research date: 11 November 2026

Repository: SadamAnjaneyulu/Algo-8-Reverse-Engineer — PRs #1–#6 (all open drafts)

Report branch: docs/algo8-complete-research-report

Documentation of completed research only: no new experiment was run for this report. Every figure is plotted from a committed result file and every number cites its source. Results from different research lineages are reported separately. All trading days in the sample were previously inspected; no finding is out-of-sample confirmation.

Contents

1. Executive summary 4
2. Data inventory and quality 6
2.1 Broker trade export 6
2.2 Market data (Breeze 1-minute) 7
2.3 What the data can and cannot establish 7
3. Reconstructed Algo 8 trading architecture 9
4. Exit-engine reverse engineering 11
4.1 Exit taxonomy 11
4.2 Individual lot stop 11
4.3 Side-level and basket-level exits 13
4.4 Cycle-level stop 14
4.5 Clock-B actions 14
4.6 Carry-forward risk (“4 % SL”) 14
4.7 Replacement and re-entry (exit-engine view) 15
4.8 Exit-engine models and replay (PR #3) 15
5. Allocation reconstruction (PR #5) 17
5.1 Allocation hypotheses 17
5.2 Slot-, chain- and lot-level ownership tests 17
5.3 Individual-lot attribution 17
5.4 Buy-back-to-resale event linking (phase 3) 18
5.5 Competing antecedents and ownership ambiguity 18
5.6 Inventory reconciliation — what “zero mismatch” means 19
6. ML decision engine (PR #1, lineage X) 20
7. OHLCV research foundation (PR #6) 22
8. Key findings and unknowns 23
9. Alpha research: what we know and what we don’t 26
10. Limitations and research risks 28
11. Current status and next research priorities 29
12. Technical appendix 30
12.1 PRs, branches and commits 30
12.2 Key files 30
12.3 Important statistics 31
12.4 Definitions 31

Repository: SadamAnjaneyulu/Algo-8-Reverse-Engineer · **Report branch:** `docs/algo8-complete-research-report` (stacked on PR #6) · **Research date:** 11 November 2026

Scope. This report documents research already completed in PRs #1–#6. It is an audit and write-up; **no new experiment was run for it.** Every number below is taken from a committed result file, a findings document, or a reviewed PR artifact, and the source is cited. Numbers from different research lineages are never placed in one comparison without an explanation of how they differ.

How to read the evidence labels.

Label	Meaning
Confirmed observation	Directly visible in the trade export (counts, timestamps, tags); not a claim about intent.
Strong evidence	Repeated, consistent pattern across splits and instruments; the exact rule is still not identified.
Partial reconstruction	Part of the behaviour (e.g. timing or quantity) is reproducible; the trigger is not.
Unresolved	Not identified in the available data. This does **not** mean the strategy has no such rule.
Rejected hypothesis	Tested and contradicted by the data.

No finding in this report is out-of-sample confirmation: **every trading day in the sample has already been inspected** (§10).

1. Executive summary

Objective. Reverse-engineer "Algo 8", a NIFTY / SENSEX weekly-options strategy run for client GKOHNP140, from its broker trade export and 1-minute market data. The supplied description (`Info.txt`) is three sentences: a *BTST algorithm with a carry-forward SL of 4 % on positions carried to expiry and 1 % on the entire capital on expiry day*, capital ₹2 crore.

What Algo 8 appears to do. It is a weekly **short-option premium-carry engine**:

- Each weekly expiry is one **cycle**: the pre-expiry session (**D-1**) and the expiry session (**D0**).
- From 09:16 on D-1 it sells at-the-money (ATM) calls and puts in one-lot child orders (65 NIFTY / 20 SENSEX), buys far out-of-the-money **wings** as protection, and adds more short straddles at **17 fixed clock windows**.
- It carries the hedged book overnight into expiry day, then manages it with per-lot buy-backs, side-level and basket reductions, and a **Clock-B** rebalancer that buys back one lot and re-sells near ATM.
- It liquidates on a **cycle-level loss limit** (≈ −₹1.30–1.35 lakh mark-to-market) or flattens late on expiry day (15:14 in the later period).

Data. A fills-only broker export (41,421 fills; 39,874 orders; 35 cycles, 23 Jun – 22 Oct 2026), and Breeze 1-minute candles: index bars for 85 sessions and option bars for the 3,243 contract-days Algo 8 traded. There is no order book, bid/ask, full option chain, decision log, allocation id or original configuration.

Method. Two independent research lineages decoded the export, validated the decode against market data, labelled every order, and tested candidate rules and machine-learning models on **chronological, whole-cycle splits** (DEV 18 / VAL 8 / TEST 9 cycles in lineage Y; six expanding folds in lineage X). Successive passes:

- an audit that corrected a data defect;
- an allocation-level study;
- a causal OHLCV feature library for indicator and alpha work.

Most important discoveries.

1. **The structure is well established; the decision logic largely is not.** The schedule (544 / 544 pre-expiry windows), opening book, ATM strike rule, wing budget, base unit, order pricing (limit = LTP × 1.15 / 0.85) and cycle stop are reproducible. Exact triggers for reductions, extra entries, re-entries and Clock-B actions are not.
2. **A data defect was found and corrected.** Column 4 ("order id") is reused across days. Grouping by it merges 4,840 real orders into 2,224 groups and silently drops 2,616 orders (6.6 %). All later work uses column 163 (`UniqueOrderIdentifier`). The first exit-engine results were withdrawn and rebuilt.
3. **A per-lot stop exists on D-1 but no single rule explains it.**
   - After a contract's minute high crosses **1.7×** its cheapest open entry, 88 % of crossings are followed by a buy-back within a minute.
   - About **95 %** of stop buy-backs occur when the algo's own quote is already ≥ **1.5×** the lowest open entry.
   - But crossing 1.5× is not sufficient: no tested reference price and constant reaches 80 % precision and recall (0 of 18 rules).
4. **No model in either lineage reproduces the decisions.** Best exit-engine classifier on TEST: lot stop precision 0.27–0.37, recall 0.52–0.68 (PR #3). The ML decision engine met its 75 % standard on 0 of 20 tasks (PR #1). Autonomous replay matches about a quarter of actual exits at best.
5. **No independent alpha is demonstrated.**
   - Algo 8's gross P&L (+₹15.5 L over 30 clean cycles, lineage Y) is the carry of options sold, mostly calls in a falling market.
   - There is no measurable timing, sizing or directional skill.
   - The market ATM-straddle premium is not statistically significant in this sample.

- A searched candidate (BTST afternoon short straddle) did not pass the reality check (p = 0.104) and is frozen for an out-of-sample test that has not yet been run.

Biggest unresolved questions. Exact entry and extra-entry conditions; the exact stop reference and why most crossings are not stopped; Clock-B act / no-act; replacement strike and quantity; allocation ownership; dynamic sizing; hedge-adjustment logic; the meaning of the “4 %” and “1 %” rules; and the short sales seen just after some cycle stops.

Have we identified the original decision engine? No. An approximate **behavioural replica** can be built (calendar, clocks, ATM strikes, wings, base units, a stop band, the cycle stop, the late flatten). The readiness index is 34 %: a subjective, unweighted score over 19 components, **none of them exact** (§8).

Have we demonstrated independent, profitable alpha? No. The only pre-registered candidate awaits its locked out-of-sample test on data not yet fetched (§9).

2. Data inventory and quality

2.1 Broker trade export

Item	Value	Source
File	`Trades_GKOHNP140.csv` (on `main`), 167 / 168 columns, no header (names inferred from Algo 7's export, which has one)	`audit/ALGO8_DATA_GAPS_AND_REQUIREMENTS.md`
Fill rows	**41,421**	`analysis/exit_engine/EXIT_ENGINE_FINDINGS.md` §2
Orders (column-163 key)	**39,874**; 16,733 closing orders (11,007 short buy-backs = 15,327 lots; 5,726 long closes)	same
Cycles	**35** (18 NIFTY, 17 SENSEX), sessions 23 Jun – 22 Oct 2026; 65 underlying-sessions	register DAT-01; `ohlcv_research/data_manifest.md`
Content	Fills only: fill time (1 s), order time (minute), side, contract, limit and fill price, quantities, validity, status, **column 163 order tag**, column 144 Algold	data gaps doc
Not present	Unfilled, cancelled, rejected or modified orders; acknowledgements; allocation / parent ids; decision reasons	same
Known gaps	Four log gaps (10 Aug, 1 Sep, 14 Sep, 29 Sep) and three D0-only cycles	same

Order identity (the correction). Column 4 ("order id") is **reused across days**:

- Grouping by it merges 4,840 real orders into 2,224 groups, 2,200 of which span different cycles.
- `groupby(oid).first()` drops 2,616 orders (6.6 %).
- Column 163 is unique per order: each tag maps to exactly one date, underlying, expiry, right, strike, side and module, and all fills of a tag finish within 57 s (0 violations over 39,874 tags).

Sources: `order_key_audit.csv`, register DAT-01 / DAT-02.

- **Impact:** six older lineage-Y scripts were re-run in a sandbox with the correct key, and no headline conclusion changed.
- **One older result could not be re-run:** the P4 strike-choice model (68.5 % top-1). It remains **unverified** (DAT-03).
- **Lineage X** was not affected.

Decoded order semantics (register DAT-05 to DAT-07, Confirmed observations):

Element	Meaning
Tag prefix	numeric = body (strategy orders); `h_` = wing hedge (IOC); `v_` = overnight lump hedge; `s_` = stop-loss module; `m_` = manual. Non-numeric tags embed a microsecond decision timestamp.
Pricing	Every order is a marketable limit at LTP × 1.15 (buy) / × 0.85 (sell). So limit ÷ 1.15 or ÷ 0.85 recovers **the price the algo saw** when it decided (the "recovered quote").

Element	Meaning
AlgoId	One per underlying book (99999 NIFTY, 1E+16 SENSEX), a platform risk actor (152432) and a separate SENSEX hedge app (2.18811E+15). AlgoIds do not identify sub-strategies.

Book reconciliation. A unit-lot replay of the fills reproduces every contract's net position. Cycle MTM from the lot book equals an independent fills-plus-candles MTM (median difference ₹0; DAT-08). Lineage X's ledger reconciles independently (252 tests, DAT-09).

2.2 Market data (Breeze 1-minute)

Audited in PR #6 Phase 0 (`ohlcv_research/data_manifest.md`, `results/data_audit_report.txt`).

Item	Index bars (NIFTY, SENSEX)	Option bars
Coverage	85 sessions per index, 22 Jun – 22 Oct 2026; identical dates	3,243 contract-days, 23 Jun – 22 Oct 2026; only D-1 and D0 of each expiry
Rows	63,678	1,230,065 (1,210,547 in session)
Duplicates	0	0
Missing bars	72 minutes, all in 12–22 Oct (15:16–15:19, sometimes 15:29)	95.2 % of contract-days have all 375 bars; 97.8 % ≥ 370; 28 have < 300
Extra bars	none	19,518 post-close bars (15:30–15:39), dropped
OHLC sanity	0 inconsistent rows	0 inconsistent rows; every close on the ₹0.05 tick
Volume	**Invalid: zero in every row**	Valid: 100 % multiples of the lot size; ≥ Algo 8's own filled quantity in 99.99 % of 24,898 traded contract-minutes
Stale prices	0.9 % zero-range bars	2.3 % zero-volume bars (99 % flat = carried price); 5.5 % flat bars

Timestamp convention, confirmed against the ledger. Bars are labelled by their **start**. 97.3 % of Algo 8 fills lie inside [low, high] of the bar labelled with the fill's own minute, against 61.3 % for the next bar and 58.7 % for the previous one.

Ledger ↔ market mapping. 100 % of orders have candles for their contract-day, and all 65 ledger sessions have index bars.

2.3 What the data can and cannot establish

Can establish	Cannot establish
What was traded, when (fills to the second; body-order decisions to the minute), at what price, in which module	Why: no decision log, no configuration, no allocation id
The price the algo saw at each decision (recovered quote)	Prices between its decisions at sub-minute resolution (bars are 1-minute)
Position and P&L at every point (exact replay)	Which lot a buy-back closed (lots in a contract are fungible; ownership is a convention)
Behaviour around the traded strikes	Anything about strikes Algo 8 did not trade (no full chain), true implied volatility, bid/ask, order-book conditions
Associations in June–October 2026	Whether any rule holds outside this sample (no untouched data)

ATM straddle coverage (both legs of the nearest strike):

- D-1: 99.8–99.9 % of minutes.

- D0: only 91 % (NIFTY) and 84 % (SENSEX), and near 0 % on the worst days, when spot left Algo 8's strike set.

3. Reconstructed Algo 8 trading architecture

Each row says what is **observed** in the export and what is **inferred** as a rule. Sources: `audit/ALGO8_STRATEGY_SPECIFICATION.md` and the register `audit/ALGO8_MASTER_FINDINGS_REGISTER.md` (IDs in brackets).

Component	Observed	Inferred rule and status
Weekly cycle	One cycle per (underlying, weekly expiry): NIFTY Mon / Tue, SENSEX Wed / Thu; one underlying per day (STR-01)	Confirmed observation
D-1 / D0	D-1 builds the book; D0 manages and liquidates it	Confirmed observation
Opening entries	First paired CE / PE short sale at **09:16** on D-1 (ENT-01); opening book matches jointly in ≈ 28 / 32 sessions	Strong evidence
ATM strike	Opening strike = nearest ATM to the completed 09:15 close, same for CE and PE (31 / 31) (STK-01). Later scheduled shorts within one strike of the moving ATM 97.4 % NIFTY / 94.6 % SENSEX; exactly ATM only ≈ 70 % / 68 % (STK-02)	Opening: strong evidence. Later: partial reconstruction. Off-grid "one strike toward the move" (STK-03 / 04): hypothesis; the 68.5 % model is unverified
Short-option legs	Book is a superposition of many one-lot allocations across neighbouring strikes, not one four-leg basket (STR-02). Old strikes are kept, not re-centred (MGT-01)	Confirmed observation
Protective wings	Seed wings NIFTY ≈ ± 2,000 pts; SENSEX ≈ ± 4,000 before 16 Sep, ± 2,000 after (HDG-01). Routine wing = highest OTM premium ≤ 2.5 % of the ATM leg (exact strike ≈ 50 %, within one strike 92–95 %) (HDG-02). Wings bought within 5 s of most new shorts (HDG-03). Carry books keep long ≥ short per right (HDG-05)	Strong evidence (approximate selector)
Quantity / lots	Child-order base unit 65 / 20 (SIZ-01); opening 3 units per role (SIZ-02); a modal ladder 3, 2, 3, 1, … shared across indices (SIZ-03)	Base unit: confirmed. Ladder: partial (≈ half of later slots). Sizing formula vs capital: unresolved (SIZ-05)
Scheduled additions	**17 pre-expiry windows** (09:30, 09:45 … 14:30; window T − 2 s … T + 3 min) contain fresh CE / PE shorts in 544 / 544 cases (ENT-02). On D0, fresh shorts continue while the book is active (95.1 % / 100 %, but given the actual book; ENT-03)	D-1 times: confirmed observation. D0: strong evidence, conditional on the actual book
Extra entries	Off-schedule sales on both days (ENT-04)	Trigger **unresolved**. Lineage Y timer model ranks minutes (AUC 0.887); lineage X rule search catches 13 / 93 events
Reductions / buy-backs	11,007 short buy-backs classified into 11 categories plus an unexplained class of 0.6 % (EXT-01)	Classification: confirmed. Triggers: §4

Component	Observed	Inferred rule and status
Clock-B rebalancing	D0 ticks at :14 / :29 / :44 / :59 from 09:29 (MGT-02). 42–47 % of ticks act; one lot per contract (MGT-03 / 04)	Timing: strong evidence. Act / no-act and contract choice: unresolved
Replacement / re-entry	Re-sales within ≈ 60 s of buy-backs are 3–4× more frequent than chance after Clock-B rebalances and lot stops (§5.4)	Association: strong evidence. Strike, quantity and trigger rule: unresolved
Stop-loss and liquidation	Per-lot stop on D-1 near 1.6–1.7× entry, one lot at a time (EXT-02 / 03). Cycle stop: the `s_` module liquidates at cumulative cycle MTM ≈ −₹1.30–1.35 L (executed −₹1.44 L on average because of 1–20 s latency; 12 stop cycles) (RSK-01)	Strong evidence (levels bracketed, not exact)
Final flattening	Short book closed in the **15:14** minute in 9 / 10 late closures after 3 Aug (EXT-08); earlier period more spread	Strong evidence for the later period; universal rule unresolved
Overnight lump hedge (`v_`)	On the heavier short side (29 / 31), 0.9–2.3 % OTM (SIZ-04)	Side / OTM: strong evidence. Size: unresolved (models worse than the naive median)

Indicators. PR #2 tested 75 causal index indicators, 14 option measures and crossover events against 8 decisions, with date-shuffled nulls and FDR control. **No classical technical or option indicator is identifiably used** (STR-03). The algo reacts to recent price movement and price location, not to a computed RSI or EMA (`FINDINGS.md` §0c).

4. Exit-engine reverse engineering

Main source: PR #3, `analysis/exit_engine/EXIT_ENGINE_FINDINGS.md`, rebuilt on the corrected order key. Split: whole cycles, chronological, **DEV 18 / VAL 8 / TEST 9**. Models are fitted on DEV, thresholds set on VAL, and TEST is scored once. TEST is not an unseen period: hand-derived constants were first seen on all 35 cycles.

Five measures that must not be mixed (PR #4 §7.0). A number from one row says nothing about another.

#	Measure	What it is
1	Exit-event association	Any buy-back of the contract within *W* minutes after a price crossing
2	Stop-consistent classification	A non-scheduled buy-back whose own quote is still beyond the threshold, quantity ≤ lots crossed
3	ML event prediction	Precision / recall / PR-AUC of a classifier on contract-minute rows
4	Autonomous portfolio replay	A model runs its own book; exit match within ± 2 min
5	Replay P&L	Cash result of the replay under a fill assumption

4.1 Exit taxonomy

Share of the 15,327 short buy-back lots (PR #3 §3; a labelling, **not** a recovery of triggers):

Category	Lot share	Category	Lot share
Individual lot stop (quote ≥ 1.45× cheapest open entry)	22.5 %	Final flatten (D0 ≥ 15:10)	13.1 %
Same-contract partial buy-back	5.3 %	Clock B without replacement	2.7 %
Side-level reduction (≥ 2 contracts, one side)	5.7 %	Expiry-day salvage (≤ ₹1 or ≤ 5 % of entry)	13.5 %
Basket-level reduction (both sides)	11.8 %	Clock-B rebalance / grid rebalance	4.9 % / 4.6 %
Cycle stop & liquidation	13.8 %	Platform / manual	1.7 %
Trailing stop / target	**no rule found**	Unexplained	**0.6 %**

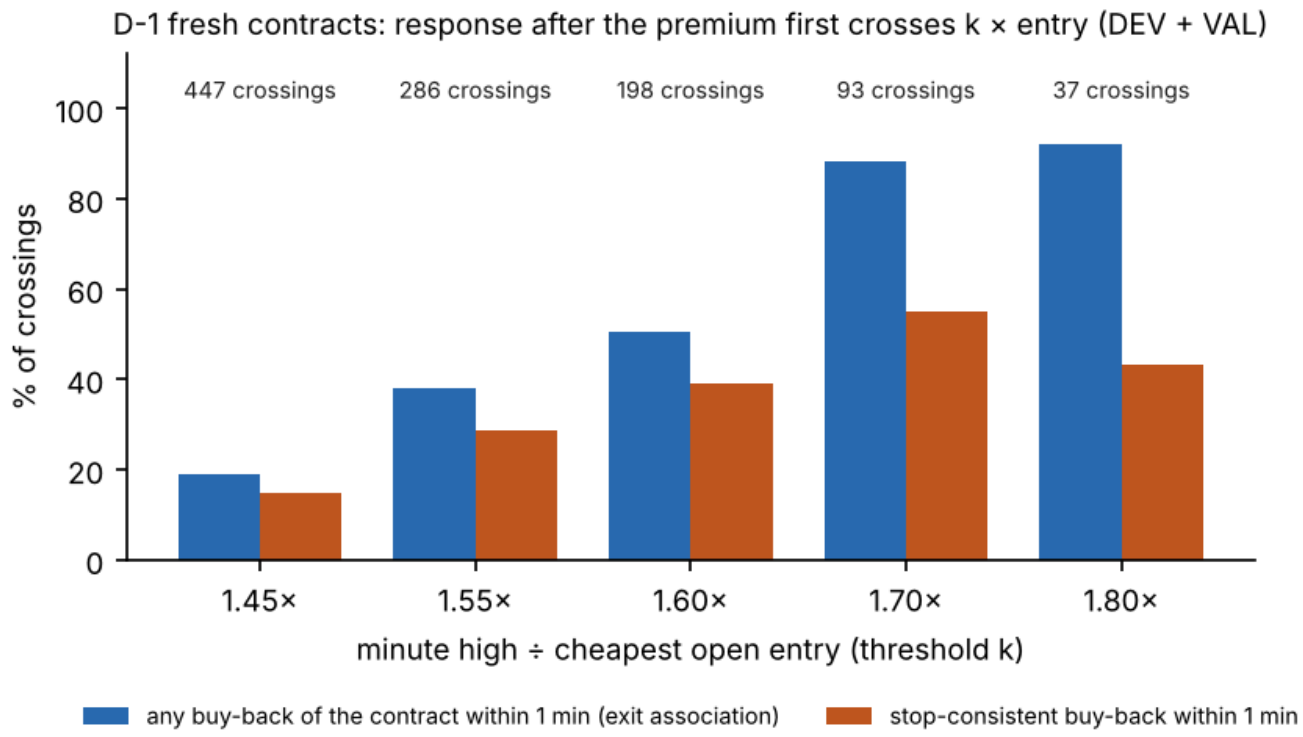
The lot-stop label itself uses a 1.45× threshold, so its zero hazard below 1.45× is by construction. The first-passage tests below are the independent check.

4.2 Individual lot stop

Hazard step (D-1). Per-minute stop hazard by minute high ÷ cheapest entry (PR #3 §4):

Ratio band	1.45–1.55	1.55–1.7	1.7–2.0	2.0–3.0
D-1	1.5 %	10.5 %	52 %	75 %
D0	2.7 %	6.9 %	12 %	9 %

First passage (DEV + VAL, D-1 fresh contracts; `first_passage_response_curve.csv`):



D-1 first-passage response by threshold. Source: PR #3 first_passage_response_curve.csv (D-1 fresh, 1-minute window).

k	Crossings	Any buy-back ≤ 1 min	Stop-consistent ≤ 1 min
1.45	447	19.0 %	15.0 %
1.55	286	38.1 %	28.7 %
1.60	198	50.5 %	38.9 %
1.70	93	**88.2 %**	54.8 %
1.80	37	91.9 %	43.2 %

At 1.7× the 1-minute association is 0.88 / 0.88 / 0.76 on DEV / VAL / TEST.

- **One lot at a time:** the stop closes exactly one lot in 79–100 % of cases.
- **Coverage is limited:** the best single threshold covers ≈ 45 % of stop-like exits on D-1.
- **D0 is weak:** carried lots show no per-lot stop at all (hazard flat at ≈ 2–3 % per minute from 1.04× to 3×).
- **No trailing stop or points floor:** `entry + max(a·entry, b)` is no better than the pure ratio.

The stop constant (PR #5 phase 2; pre-registered).

Design:

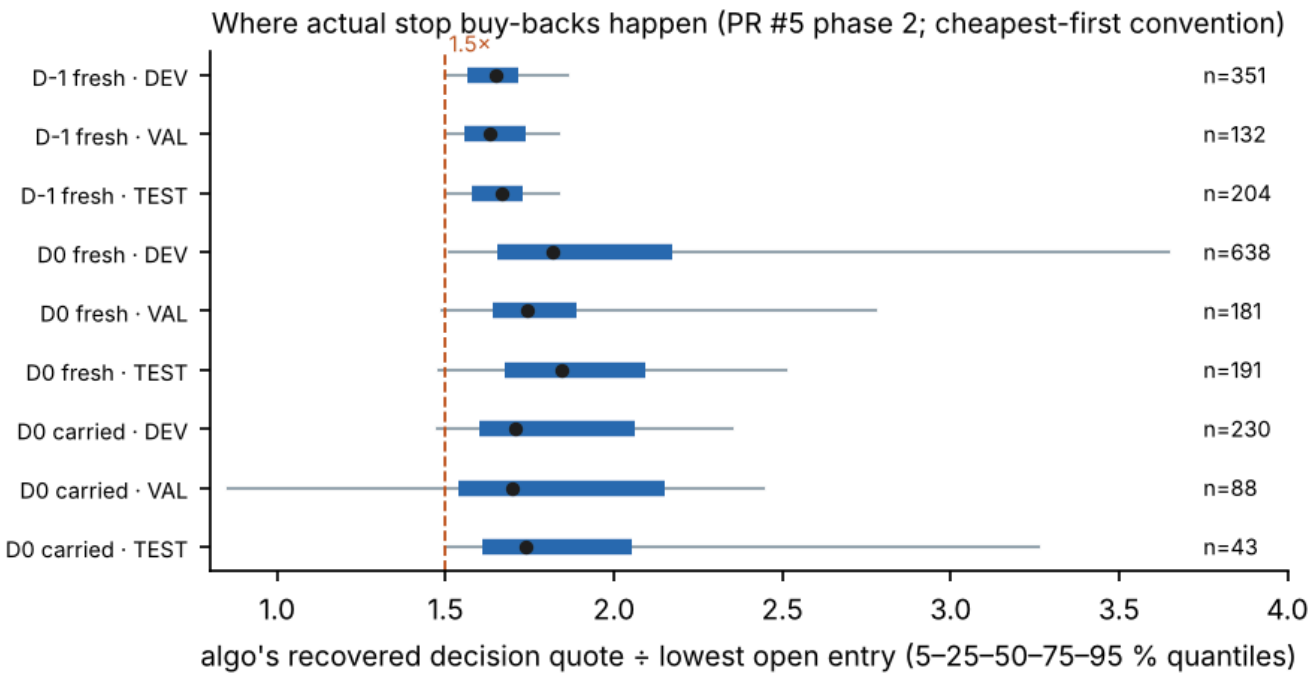
- **References:** six (own entry, cheapest open entry, slot-allocation average, chain-allocation average, lot-weighted and order-average contract averages).
- **Grid and conventions:** *k* = 1.40–2.00, three ownership conventions, three position segments.
- **Matching:** one buy-back to at most one event, confirmed with the algo's own recovered quote.
- **Selection:** *k* chosen on DEV only.

Result:

- **Verdict: UNKNOWN retained.** 0 of 18 (convention × reference) global rules reach 80 % precision and recall on VAL and TEST (`stop_ref_verdict.csv`).
- **Best rule:** own entry, *k* = 1.55. TEST precision **0.27** [0.21, 0.32], recall **0.63** [0.58, 0.68] (90 % cycle bootstrap); 1,011 of 1,381 events unanswered.

- **Ceiling:** even choosing *k* on each split's own data, the best min(precision, recall) is 0.52 (0.66 with looser matching).

What the data do support (`stop_ref_exit_ratio_quantiles.csv`):



Recovered quote ÷ lowest open entry at actual stop buy-backs. Source: PR #5 stop_ref_exit_ratio_quantiles.csv.

- **Necessary lower edge:** the 5th percentile of the algo's recovered quote ÷ lowest open entry at actual stop buy-backs is 1.50 / 1.51 / 1.51 (D-1, DEV / VAL / TEST), with median 1.63–1.67. About 95 % of stop buy-backs happen at ≥ 1.5×.
- **Not sufficient:** most 1.5× crossings are not stopped within minutes. Matched responses are spread over 0–5 minutes: 21 % in the crossing minute, 38 % the next minute, 41 % later (`stop_lag.csv`).

Earlier conflicting statistic, reconciled (PR #4 §7A). PR #2 reported "74 % of D-1 lots stopped within 2 min at 1.55× own entry".

- **It reproduces on clean data under its own definition:** 74.1 % of 1,045 lot events.
- **It is inflated by composition:** 22 % of events had an exit *before* the trigger, 851 of 1,045 are follow-on lots of a contract already crossed, and 88 fire when no lot is still open.
- **Population-matched comparison:** on the 151 contracts common to both studies, first crossings give PR #2 58.3 % vs PR #3 67.5 % within 2 minutes.
- **Status:** the 1.55× claim is **rejected as stated**. The band 1.6–1.7× is strong evidence; the exact constant is unresolved.

4.3 Side-level and basket-level exits

Item	Side-level	Basket
Book-minute LightGBM, VAL (PR #3 §6)	AUC 0.87 (base rate 3.5 %, AP 0.25)	AUC 0.86 (2.2 %, AP 0.15)
Clock + inventory baseline AUC	0.80	0.85
Contract-minute AUC gain from spot move (T4)	**+0.21**	+0.03
Depth-3 tree VAL precision	0.11	0.05
TEST classifier (LightGBM / CatBoost P / R)	0.21 / 0.40, 0.22 / 0.39	0.19 / 0.28, 0.14 / 0.38

Item	Side-level	Basket
Which contracts are closed (hit@1 vs random, VAL)	largest exposure 0.77 vs 0.47	0.58 vs 0.35

- **Side-level reductions** respond to adverse spot moves against the short side (strong evidence of association).
- **Basket reductions** are mostly timing-driven.
- **Neither trigger is recovered.** Status: unresolved.

4.4 Cycle-level stop

- **Mechanism (RSK-01, strong evidence):** the `s_` module liquidates the whole cycle when cumulative LTP mark-to-market (D-1 + D0, body + hedges) reaches ≈ −₹1.30–1.35 L. It is size-independent (56–209 short lots) and fires on both days.
- **Execution:** executed MTM averages −₹1.44 L (SD ₹6.8 K over 10 non-gap stops) because of 1–20 s check latency. There were two gap-through stops (−₹2.1 L, −₹2.5 L).
- **First-passage check:** every minute-close crossing of −₹1.25 L to −₹1.5 L ended in a stop, so there are no false triggers at that level (PR #3 §5).
- **"1 % of capital on expiry day"** (₹2 L at ₹2 Cr) is **not reproduced**: the observed limit is ≈ −₹1.3 L and fires on D-1 too (RSK-03, rejected as "₹2 L expiry stop").
- **Open item (flagged, PR #5 ALC-09):** in 7 of 12 cycle-stop cycles, short-opening fills arrive within ≈ 140 s after the liquidation starts (603 lots link to a liquidation buy-back up to 38 s earlier). They may be in-flight orders; the export has no order lifecycle. This sits uneasily with the register's "no new entries afterwards" wording and is **unresolved**.

4.5 Clock-B actions

- **Timing and quantity:**
  - A tick acts in 46.5 % (DEV) / 42.3 % (VAL) of D0 ticks (MGT-03).
  - It buys back **one lot** (89–99 %) and re-sells the same lots a median 28 s later (PR #3 §6).
- **Act / no-act:** partly predictable (LightGBM VAL AUC 0.78; depth-3 tree precision 0.77, recall 0.55; strongest signals time-to-expiry, time since last trade, theta exposure) but **unresolved**.
- **Contract choice:**
  - Best single criterion "most lots" 0.66 hit@1 vs 0.41 random (VAL).
  - On TEST: LightGBM top-1 0.56, CatBoost 0.48, vs 0.35 random (59 tick-sides).
  - Status: unresolved.
- **"Re-sold at ATM" holds only loosely** (PR #5 phase 3): 66 of 203 re-sold lots within 60 s are at 0 strike steps from ATM; the rest are 1–3 steps either side.

4.6 Carry-forward risk ("4 % SL")

Six interpretations were tested, and none separates trigger from non-trigger cases on DEV, VAL and TEST (PR #3 §7; RSK-02):

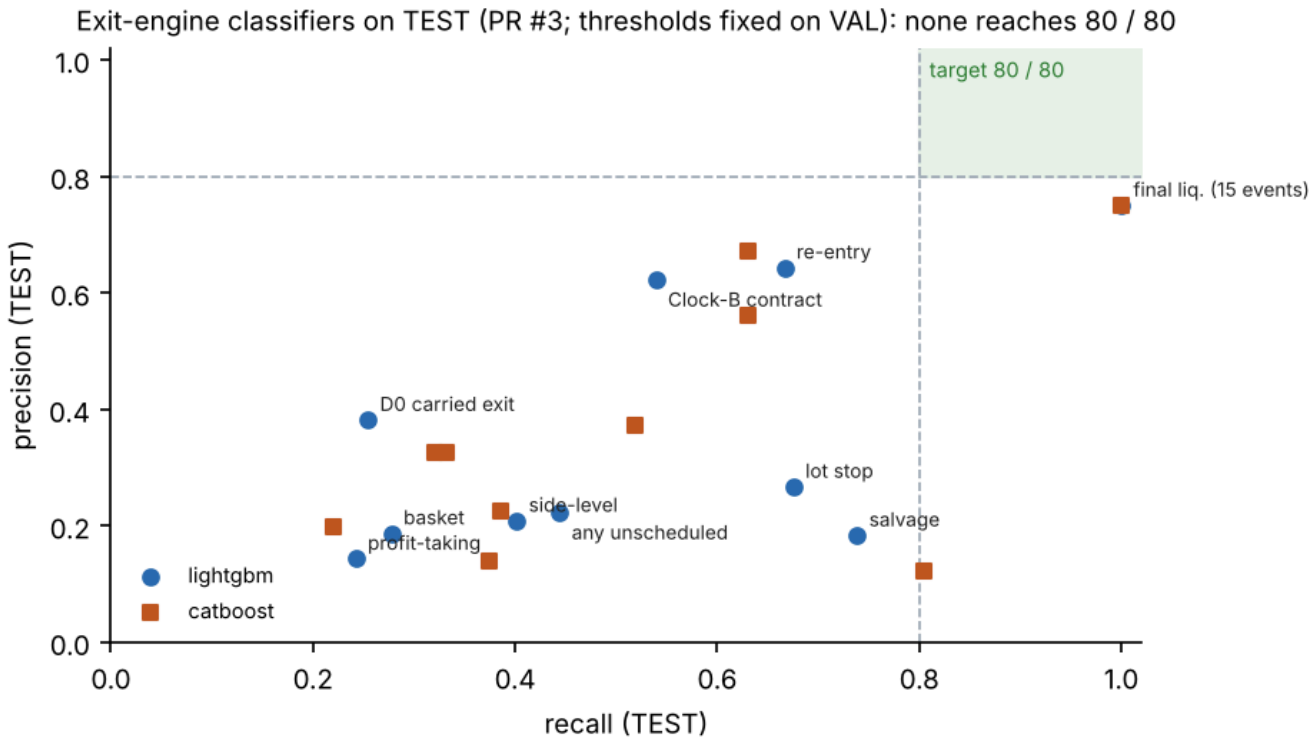
- carried premium ≥ 1.04 × D-1 close;
- book value ≥ 1.04 ×;
- loss ≥ 4 % of ₹2 Cr or ₹1 Cr;
- D0 loss ≥ 4 % of |MTM at D-1 close|;
- spot move ≥ 4 %;
- drawdown ≥ 4 % of premium collected.

The meaning of "4 %" is **unresolved**. Carried lots also show no per-lot premium or spot stop.

4.7 Replacement and re-entry (exit-engine view)

The re-entry classifier (same-side sale within 60 s of a buy-back episode) has the only clear lift among exit classes, but remains below target: TEST precision 0.64–0.67, recall 0.63–0.67 against a 44 % base rate. The allocation study (§5.4) quantifies the association more carefully.

4.8 Exit-engine models and replay (PR #3)



Exit-engine classifiers on TEST. Source: PR #3 exit_model_metrics.csv.

Model (TEST)	Events (base rate)	LightGBM P / R	CatBoost P / R
Lot stop	339 (0.6 %)	0.27 / 0.68	0.37 / 0.52
Any unscheduled buy-back (label-free)	788 (1.4 %)	0.22 / 0.44	0.33 / 0.33
Side-level reduction	122 (1.3 %)	0.21 / 0.40	0.22 / 0.39
Basket reduction	72 (1.5 %)	0.19 / 0.28	0.14 / 0.38
D0 carried-position exit	228 (1.7 %)	0.38 / 0.25	0.32 / 0.32
Salvage	153 (0.6 %)	0.18 / 0.74	0.12 / 0.80
Re-entry after a buy-back	192 (44 %)	0.64 / 0.67	0.67 / 0.63
Final liquidation	**15** (12.5 %)	0.75 / 1.00	0.75 / 1.00 (too few events)
Multi-class controller	macro-F1 0.17		

Robustness and ablation.

- **Bootstrap:** 90 % cycle-bootstrap intervals on TEST for the lot stop are precision 0.25–0.30 and recall 0.63–0.74.
- **D-1 segment:** precision 0.34, recall 0.84; the model over-fires where the stop is crisp.
- **Ablation:** removing the premium ratio cuts TEST average precision from 0.39 to 0.27, and removing the spot move to 0.33. Removing Greeks, IV, inventory, timing or clock costs nothing.
- **Interpretation:** the classifiers learn the same price-move signal, not a hidden rule.

Autonomous replay (TEST, 9 cycles; actual TEST P&L +₹7.5 L):

Fill assumption	Mode A (actual entries) exit-match P / R; replay P&L	Mode B (fully autonomous) P / R; replay P&L
Minute open	0.40 / 0.24; +₹4.6 L	0.37 / 0.12; −₹3.5 L
One-minute delay	0.40 / 0.22; +₹5.7 L	0.36 / 0.12; −₹3.5 L
Adverse (buys at high, sells at low)	0.47 / 0.16; −₹10.3 L	0.38 / 0.12; −₹6.1 L

- **Fills don't explain the low fidelity:** which exits match barely moves across fill assumptions, so this is a decision problem, not an execution problem.
- **Replay P&L is not evidence:** it swings sign with fills, and Mode A's P&L similarity comes from the actual entries it is given. **Replay P&L is not evidence of reconstruction.**

Why exact triggers remain unknown. The data show *that* the algo acts near certain price levels and clocks, but not:

- which allocation it was managing;
- when it checked;
- what else it evaluated.

Every model reaches the same ceiling with the same features. The missing ingredient is information (decision logs, allocation ids, order lifecycle), not model capacity.

5. Allocation reconstruction (PR #5)

Sources: `analysis/allocation/ALLOCATION_FINDINGS.md` (phases 1–2) and `analysis/allocation/REPLACEMENT_REENTRY_FINDINGS.md` (phase 3).

5.1 Allocation hypotheses

Allocation ids are not in the data, so "allocation" is a hypothesis. Three definitions were built from the same clean orders and reported separately:

Definition	Rule	Short allocations	Mean lots (single-lot share)
Slot	One per (cycle, date, schedule slot, right); off-grid sales form "extra" allocations per sale burst	6,545	2.26 (56 %)
Chain	Lineage Y's timer-phase chains, re-run on clean bursts (gap > 4 s, which reproduces the legacy 5,481 bursts exactly)	4,355	3.40 (63 %)
Lot	Every unit lot is its own allocation	—	1

Which lot a buy-back closes is unobservable. Every result is repeated under **cheapest-first, FIFO and LIFO** ownership conventions.

5.2 Slot-, chain- and lot-level ownership tests

Test	Result	Reading
Contract chosen at D-1 stop buy-backs (691 orders; random hit@1 0.115)	Lowest individual entry 0.560; slot average 0.551; chain average 0.556; **contract average 0.493** (paired gap 6.7 points, 90 % CI [3.4, 9.9])	The stop references an **individual entry**, not the contract-average position. Strong evidence
Reference sharpness (AP of the raw ratio, D-1)	Lowest entry 0.36 / 0.24 / 0.39 (DEV / VAL / TEST) ≥ slot ≥ chain > contract average 0.22 / 0.13 / 0.32	Same ordering on all splits
Allocation-sized stops	Stops close one lot; 14–19 % of D-1 stops contradict a whole-allocation close	Partly contradicted
Discriminating test (entries ≥ 10 % below the average)	Contract average rejected on D0, unsupported on D-1; slot / chain averages **not separable** from the individual entry	Allocation identity unresolved
Wing coupling	Released wings belong to the closed short's allocation 26.6 % (slot) / 40.6 % (chain) vs permutation nulls 17.7 % / 34.5 %	Weak association

5.3 Individual-lot attribution

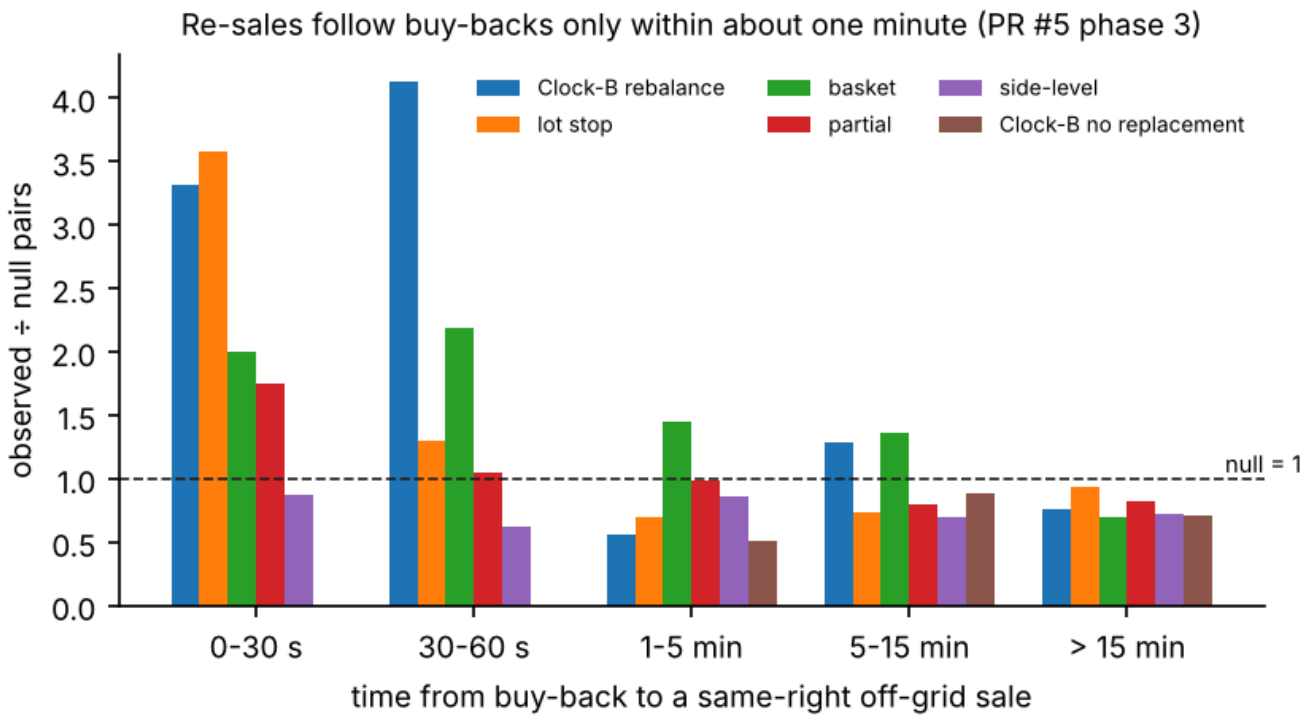
Because 56–63 % of slot / chain allocations are single lots, these tests **cannot separate "per lot" from "per small allocation"**. The data support the statement "the stop is anchored to an individual entry or a very small allocation". They do not identify which.

5.4 Buy-back-to-resale event linking (phase 3)

Linker.

- Every buy-back (11,007 orders, 15,327 lots) is linked to later same-right sales (13,410 orders, 15,312 lots).
- **Capacity:** no buy-back is matched to more lots than it closed, and no sale lot is used twice (both asserted in code). Splits are explicit rows.
- **Scheduled-window sales** are treated as independent entries.

Association vs a time-shifted null (`replacement_association_null.csv`):



Re-sale association by time window. Source: PR #5 replacement_association_null.csv. "Clock-B no replacement" has zero pairs within 60 s.

Reduction type	Observed ÷ null, 0–30 s	30–60 s	Share of lots re-sold ≤ 60 s
Clock-B rebalance	3.31	4.13	31.5 % (≈ 70 % ignoring competing buy-backs)
Lot stop	3.58	1.30	4.1 %
Basket reduction	2.00	2.19	18.1 %
Partial buy-back	1.76	1.05	20.0 %
Side-level reduction	0.87	0.62	6.2 %
Clock B, no replacement	0 pairs	0 pairs	0 %

- **Within about a minute:** re-sales are real and Clock-B labels behave as their names say.
- **Beyond one minute:** observed pairs are mostly at or below the null, so later "replacements" are indistinguishable from independent entries.
- **Side-level reductions** are not followed by re-sales, which is consistent with genuine de-risking.

5.5 Competing antecedents and ownership ambiguity

Of 3,611 replacement / re-entry lot pairs:

Outcome	Share
Cannot distinguish: competing buy-backs on ≥ 2 contracts within 15 min	73.1 %
Cannot distinguish: owner changes with the ownership convention	23.4 %
Slot and chain models say “new allocation”; event model says “continuation”	3.2 %
Ownership **identified**	**3.5 %**

Slot and chain models almost never treat a re-sale as continuing the old allocation (0.4–0.8 % and 2.6–3.6 %). **Allocation ownership is not identified in the available data.**

5.6 Inventory reconciliation — what “zero mismatch” means

- **The result:** all nine lifecycle ledgers (three linkage models × three conventions) reproduce the observed short inventory after every one of **39,874 orders**, with **0 mismatches** and 0 negative inventories (`state_transition_metrics.csv`).
- **What it proves:** the replay accounting, linkage bookkeeping and capacity constraints are correct, and no event is double-counted.
- **What it does not prove:** that any allocation model is right. Conservation holds **by construction**, because every model partitions the same exact lot replay. A wrong ownership story conserves inventory just as well. Zero mismatch is a software check, not evidence of reconstruction.

6. ML decision engine (PR #1, lineage X)

Kept separate from the exit-engine models: different target, dataset and evaluation (PR #4 §7C).

Item	PR #1 (lineage X)
Source	PR #1, code `research/ml_decision_engine/` at `c5c4f285`; results in GitHub Actions run 41223605040, artifact `algo8-research-report` (expires 30 Nov 2026; reviewed 9 Nov)
Dataset	**1,280,181** minute-opportunity rows; input revision `dee20138`
Target	The **next-minute execution action actually received**, at basket / side / contract level (≈ 20 tasks, e.g. `add_both`, `buy_protection`, `new_short`, `reduce_contract`, `reenter`). Decision intent is not observed
Folds	Six expanding chronological folds: train 8 → 28 cycles, each testing the next 4 cycles (24 test cycles); inner validation 2 cycles
Model families	Logistic, LightGBM, CatBoost, GAM, discrete hazard
Feature sets	clock + inventory, spot / ATR, option IV, Greeks + P&L, full
Registered acceptance	Precision ≥ 0.75 **and** recall ≥ 0.75, ≥ 30 target events, ≥ 5 cycles, ≥ 0.8 coverage (`protocol.json`, fixed before training)

Results.

- **0 of 20 action / level tasks meet the registered standard.** Best full-feature rows:

Task	Precision / recall	Note
basket `initial_entry`	0.83 / 1.00	only 19 events, below the 30-event minimum
`add_both`	0.70 / 0.62	
side `buy_protection`	0.73 / 0.59	
contract `new_short`	0.38 / 0.46	
`reduce_contract`	0.37 / 0.43	
`reenter`	0.08 / 0.22	

- **Autonomous replay is abstention-limited:** 288 rows (24 cycles × 4 roles × 3 tolerances), **0 predicted orders and 0 simulated fills**. The "final inventory reproduced" flag (True in 216 / 288 rows) describes an empty replayed book and is **not** evidence of reproduction.

Earlier lineage-X controller studies (on `research/strong-findings`; register ML-01):

- A constrained autonomous controller selected in August reached a later-period event macro-F1 of 23.3 % vs 18.2 % for a clock baseline.
- A policy / execution separation reached 24.3 %.
- An unrestricted rule search (15.9 M contexts) fit history only with hundreds of branches.

Why the ML did not recover the algorithm.

1. **The target is the received action, not the decision.** Unfilled orders, allocation ids and decision times are not in the data.
2. **Actions are rare and clock-driven.** Clock + inventory already explains most predictable variance; market features add little (PR #1's `incremental_market_feature_value.csv`).
3. **Effective sample is small:** 35 cycles, ~24 test cycles, and the two indices move together.

4. **Shared failure point:** both lineages, with different targets and models, fail in the same place. That points to missing information, not to a modelling choice.

7. OHLCV research foundation (PR #6)

Source: PR #6, folder `ohlcv_research/`. **Status: Phase 0 (data audit) and Phase 1 (feature library) are complete. Phase 2 (indicator-trigger models and alpha screening) is not complete:** no Phase-2 result is in the repository (see §11).

Component	What exists
Phase 0 data audit	`data_manifest.md`, `data_audit.py`, audit CSVs (summarised in §2.2)
Feature library	`features.py`, `OHLCV_FEATURE_CATALOG.md`: **42 market features in six groups + 4 clock fields**, every formula documented
Index features (25)	Momentum: returns 1–60 min, 5-min EMA slope, RSI, MACD histogram, ADX. Mean reversion: 60-min z-score, Bollinger %B, EMA distance. Volatility: ATR, realised and Parkinson volatility, range expansion, Bollinger width. Market structure: return from open, opening-range position / break, distance to previous-day high / low, gap, session range / position
Option features (11)	Premium returns 1 / 5 / 15 min, premium / spot, 5-min range, VWAP distance, relative volume, volume z-score, OBV imbalance, Chaikin money flow, bar age
ATM straddle features (6)	ATM strike, CE + PE premium ÷ spot, straddle returns (strike held fixed), ln(CE / PE), ATM relative volume. This is an observed premium, **not** implied volatility
Feature correlations	8 pairs with Spearman $\|\rho\| > 0.9$ and 16 > 0.8, e.g. RSI / EMA distance / EMA slope / %B at 0.86–0.98. Results can be attributed to **families**, not to a named formula
Leakage prevention	Each row keyed by decision minute *m* uses only bars completed before *m*; the forming candle is never used; 5-min bars released at their end; sessions, contracts, expiries and instruments never mix; no full-sample fitting
Causality tests	25 unit / leakage tests pass (formulas vs independent loops, truncation invariance on synthetic and real data, split boundary, future- and current-candle perturbation, isolation, post-close bars, straddle rules, staleness, loaders). **Mutation check:** 12 deliberately injected leaks, all caught (`mutation_test_report.txt`)
Proposed opportunity panels	`OPPORTUNITY_PANEL_DESIGN.md`: scheduled-entry sizing, extra entries, individual buy-backs (including a "crossed but not stopped" panel), side-level, basket, Clock-B ticks, hedge adjustments. Event counts are in `opportunity_counts.csv`
Baseline models (proposed)	B1 = clock + calendar + D-1 / D0 + inventory. B2 = B1 + premium relative to entry + cycle MTM. A market family must add information over **B2**, so the known premium-stop effect is not mistaken for an indicator
Research tracks	**Track A:** does Algo 8 use OHLCV-derived conditions? **Track B:** independent alpha (five families: option premium vs realised move, momentum, mean reversion, opening-range breakout, volatility compression). Kept separate

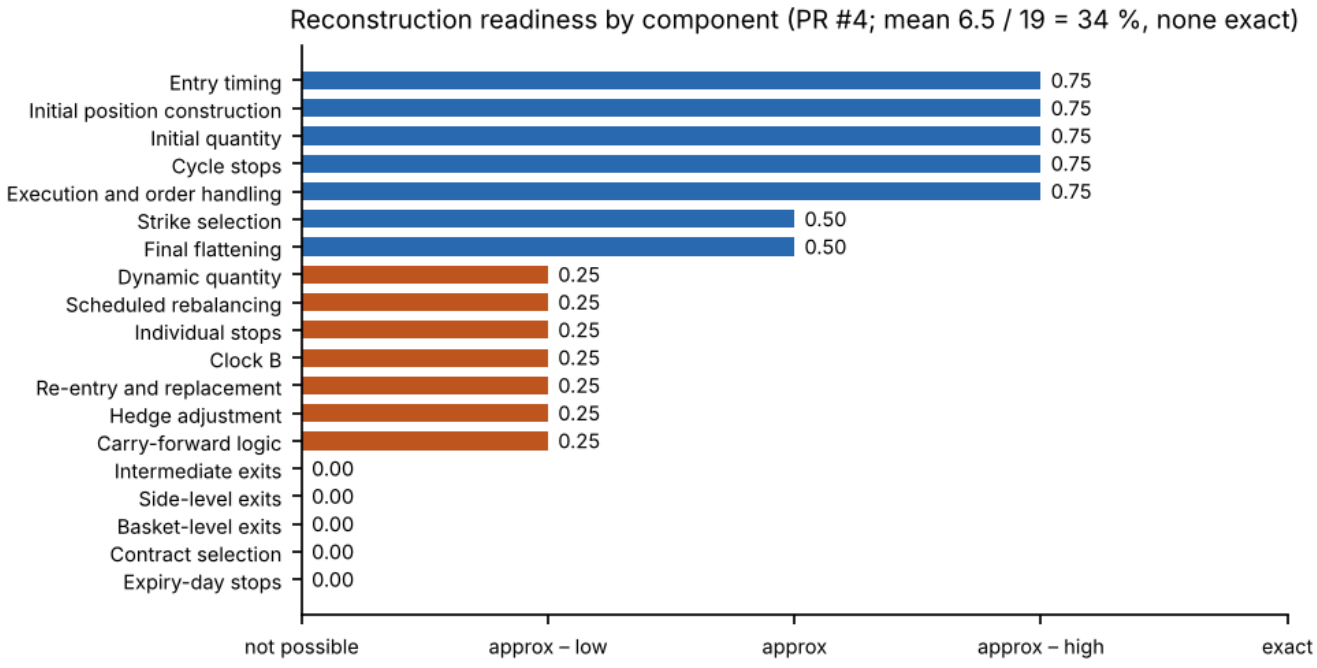
8. Key findings and unknowns

Component	Finding	Evidence	Confidence	Status
Order identity	Column 4 reused across days; column 163 unique	4,840 orders in 2,224 col-4 groups; 0 violations on col 163 (DAT-01 / 02)	High	Confirmed observation
Order pricing	Limit = LTP × 1.15 / 0.85; decision quote recoverable	DAT-05	High	Confirmed observation
Module tags	body / `h_` / `v_` / `s_` / `m_`	DAT-06	High	Confirmed observation
Cycle structure	D-1 build-up, D0 management	STR-01	High	Confirmed observation
Pre-expiry schedule	17 windows, 544 / 544	ENT-02	High (in-sample)	Confirmed observation
Opening book	09:16, ATM from 09:15 close, 3 units per role, seed wings	ENT-01, STK-01, SIZ-02, HDG-01 (≈ 28 / 32 jointly)	High	Strong evidence
Later strikes	Within one strike of ATM 95–97 %; exact ≈ 70 %	STK-02	Medium	Partial reconstruction
Wing budget	≤ 2.5 % of ATM leg premium; ± 1 strike 92–95 %	HDG-02	Medium–high	Strong evidence
Ladder quantities	Modal ladder; ≈ half of later slots match	SIZ-03	Medium	Partial reconstruction
Indicators	No classical indicator identifiably used	STR-03 (PR #2)	Medium–high	Strong evidence (negative)
D-1 lot stop	Response 88 % at 1.7×; ≈ 95 % of stops at ≥ 1.5×; one lot at a time	EXT-02 / 03; PR #5 phase 2	Medium	Strong evidence (band); exact rule unresolved
“1.55× own-entry, 74 %”	Inflated by composition	PR #4 §7A	High	Rejected hypothesis (as stated)
Single stop rule ≥ 80 / 80	0 of 18 rules	PR #5 phase 2	High	Rejected hypothesis
Trailing stop / points floor	Not better than the pure ratio	EXT-06	Medium–high	Rejected hypothesis
Cycle stop	≈ −₹1.30–1.35 L cumulative MTM, both days	RSK-01 (12 stops)	Medium–high	Strong evidence
“1 % expiry stop = ₹2 L”	Not observed	RSK-03	High	Rejected hypothesis
“4 % carry SL”	No interpretation separates triggers	RSK-02	High (negative)	Unresolved
Side-level reductions	Respond to adverse spot moves; trigger unknown	MGT-05; T4 +0.21	Medium	Partial reconstruction
Basket reductions	Timing-driven; trigger unknown	MGT-05	Medium	Unresolved
Clock B	Ticks, one lot, re-sale ≈ 28–48 s	MGT-02–04; PR #5 phase 3	Medium–high	Partial reconstruction

Component	Finding	Evidence	Confidence	Status
Stop reference level	Individual entry, not contract average	PR #5 T2 / T3	Medium	Strong evidence
Re-sale association	Real within ≈ 60 s, not beyond	PR #5 phase 3	Medium–high	Strong evidence
Allocation ownership	Identified for 3.5 % of pairs	PR #5 phase 3	High (negative)	Unresolved
Post-cycle-stop short fills	7 of 12 stop cycles	ALC-09	Low	Unresolved (flagged)
ML reconstruction	No class ≥ 80 / 80 (PR #3); 0 / 20 tasks ≥ 75 / 75 (PR #1)	PR #3, PR #1	High (negative)	Rejected hypothesis ("an ML model reproduces Algo 8")
Economic alpha	Not demonstrated	ALP-01, ALP-02	Medium–high	Unresolved / not shown

Most important unknowns (not identified in the available data):

Unknown	Best current evidence	What would resolve it
Exact entry and extra-entry conditions	Schedule known; extra entries ranked by timers (AUC 0.887) but not triggered	Decision logs or configuration
Exact stop trigger and ownership	Necessary ≥ 1.5×; band 1.6–1.7× on D-1; no rule ≥ 80 / 80	Sub-minute decision times, allocation ids
Clock-B act / no-act	AUC 0.78, tree 0.77 / 0.55	Decision logs
Replacement strike and quantity	Re-sale ≤ 60 s, mostly a different strike, not strictly ATM	Full chain, allocation ids
Allocation ownership	3.5 % identified	Allocation / parent ids
Dynamic sizing	Modal ladder only; lump size unpredictable	NAV / margin records, configuration
Hedge adjustment logic	Budget rule approximate; releases loosely tied to own allocation	Configuration, full chain
Cycle-stop behaviour and in-flight orders	Constant bracketed; post-stop fills unexplained	Order-lifecycle (submit / ack / cancel) records



Reconstruction readiness by component. Source: audit/ALGO8_RECONSTRUCTION_READINESS.md (subjective, unweighted index).

Readiness index (PR #4, `ALGO8_RECONSTRUCTION_READINESS.md`):

- **Method:** 19 components, each placed by judgement into Exact 1.0 / Approx-high 0.75 / Approx 0.5 / Approx-low 0.25 / Not possible 0, then averaged.
- **Result:** 6.5 / 19 = **34 %**; 25 % if approx-low counts 0; 74 % of components implementable at all; **0 exact**.
- **Caveat:** it is a subjective evidence index, not the share of code recovered and not a probability of replication. Phases 2–3 of PR #5 changed no score.

9. Alpha research: what we know and what we don't

Four objectives are often conflated. Success in one does **not** imply success in another.

#	Objective	Status	Why it does not carry over
1	**Reconstruct Algo 8's mechanics** (what it does)	Largely done for structure	Knowing the schedule and stop band does not tell us its decisions
2	**Recover Algo 8's exact decision logic**	**Not achieved** (0 / 19 exact; no model ≥ 80 / 80)	A faithful replica could still lose money if the original does not have an edge
3	**Discover an independent alpha** from market data	**Not demonstrated**	A statistically real effect may be too small to survive costs
4	**Build a profitable, executable strategy**	**Not built**	Requires 3, plus realistic execution, costs and capacity

What the evidence says about Algo 8's own P&L (lineage Y, PR #2 `FINDINGS.md` §0d; gross of costs; 30 complete cycles):

Component	₹ lakh
Body short sales held to expiry (selection)	**+31.1**
Management of those shorts (stops, rebalances, exits)	−5.8
Wings	−5.9
Overnight lump	−4.0
Actual total	**+15.5** (t = 1.39)

- **Skill:**
  - Timing vs selling the ATM straddle at every minute of the same session: −0.3 bp (D-1) and −3.0 bp (D0).
  - Sizing: Spearman +0.17, p = 0.34.
  - Directional forecasting: book delta vs next 15 / 60-min move, correlation −0.007 / +0.025.
  - None is detectable.
- **Market premium:** the ATM straddle sold on D-1 and held to expiry earned **+3.9 bp of spot**, 95 % CI **[−12.3, +19.3]**. It is negative after removing the best 2 sessions, so options look roughly fairly priced in this sample.
- **Direction:** the carry is directional by leg: short calls +₹53.8 L, short puts −₹22.7 L, in a window where both indices fell (NIFTY ≈ −7 %).
- **Lineage X's independent attribution** (ALP-01): marked P&L ₹18.9 L over 32 cycles, ₹4.75 L under conservative costs, −₹4.8 L under adverse execution; removing the two largest winners turns it negative.
- **Conclusion:** Algo 8 harvests short-dated option premium. Its profit in this window is consistent with selling calls into a falling market, **not** with demonstrated informational alpha.

The candidate strategy (PR #2 §0e; frozen in `alpha_freeze/`).

Item	Value
Search	4,512 structural configurations + 480 execution-stress variants on the development sample (Jun–Oct 2026, 30 cycles)
Best region	BTST: sell the ATM straddle on D-1 afternoon, 1.55× per-lot stop, exit at the D0 open
Pre-registered H1	Sell 1 ATM straddle at 13:16 on D-1, per-lot stop 1.55×, exit D0 09:16 (`alpha_freeze/PRIMARY_HYPOTHESIS.md`)
Development statistics	Searched-grid numbers: e.g. BTST median Sharpe 2.99 vs D0-only −0.55; execution-stressed top region Sharpe 5.2, PF 3.0

Item	Value
Selection-bias controls	**White reality check p = 0.104; probability of backtest overfitting (PBO) = 0.25** → not significant after accounting for the search
Freeze	SHA-256 `c0623b61…` and `21856dcb…`, commit `d695b777`; `oos_run.py` refuses to run if hashes change
Status	**Hypothesis.** The locked out-of-sample test needs 2025–26 history (not fetched; needs Breeze credentials) and confirmed lot sizes. It has **not been run**
Other	Cheap far-OTM short calls (< ₹10) were the only module whose carry CI excluded zero (+₹0.88 L, CI [+0.42, +1.37]); labelled exploratory, not pre-registered

Lineage X's documents mention a prior backtest screenshot (3 Mar 2025 – 27 May 2026: return 47.22 %, CAGR 36.88 %, Sharpe 2.19). It covers a different period, comes with no trades, and was **not reproduced**; it is not evidence here.

Track B (independent alpha from OHLCV). Five families (option premium vs realised move, momentum, mean reversion, opening-range breakout, volatility compression) are named in PR #6's research plan. A pre-registered hypothesis register was drafted for Phase 2 but exists only in an unpushed work-in-progress commit. **No screening result exists in the repository.** Nothing can be said yet about momentum, mean reversion, breakout or volatility-compression alpha in this sample.

Bottom line. No independent profitable alpha has been demonstrated. The single candidate is honestly labelled "not yet proven" by its own selection-bias tests and awaits genuinely new data.

10. Limitations and research risks

Risk	Detail	Consequence
Limited sample	35 cycles over four months (Jun–Oct 2026); 12 cycle stops; 9 TEST cycles	Wide intervals; small classes (Clock-B, side-level) cannot support strong claims
NIFTY–SENSEX correlation	Open-to-close return correlation 0.95	≈ 85 independent days, not 170 underlying-sessions
No untouched dates	Every session was inspected by at least one study; TEST is contaminated for hand-derived constants	All results are exploratory; nothing is confirmed out of sample
Missing full option chain	Only contracts Algo 8 traded (36–44 strikes per session-expiry)	Strike selection and hedge selection cannot be tested without bias; ATM coverage on D0 incomplete
OHLCV-only	No bid / ask, no order book; index volume is zero	No order-flow or liquidity features; no true implied volatility
Spread and fill uncertainty	Fills must be assumed (minute open, next open, adverse high / low)	Replay P&L swings from +₹4.6 L to –₹10.3 L on identical decisions
Cost assumptions	Turnover / per-order / per-unit cost schedules are assumptions, not verified broker charges	Net results are scenario-dependent
Minute-bar resolution	Body orders stamped to the minute; trigger times bracketed, not known	Exact trigger levels and latencies cannot be measured
Multiple testing	Thousands of configurations (e.g. 4,512) and many rule variants were examined	Selection bias; reality check and PBO show the best result is not significant
Model instability / few events	Final liquidation has 15 TEST events; Clock-B tick panels are small	Precision / recall on small classes is fragile
Label dependence	Exit categories are rule-based (e.g. 1.45× for “lot stop”)	Circularity, partly handled with label-free variants

11. Current status and next research priorities

Status (11 November 2026).

- **PRs:** six draft PRs, none merged. PRs #1–#5 are complete for their stated scope; PR #6 has completed Phases 0–1.
- **Phase 2 status:** Phase 2 (Track A indicator triggers, Track B alpha screening) was started on 11 November and then halted on request.
- **What exists of it:** only a local, unpushed work-in-progress commit on `research/algo8-phase2-indicator-alpha`:
  - opportunity-panel builder;
  - model runner;
  - screening script;
  - pre-registered Track B hypothesis register;
  - outputs for one panel only.
- **None of it is in the repository or reported here.**

Ranked next tasks (bounded; each produces a decision-relevant output):

Rank	Task	Expected output	Why it matters
1	**Run the locked out-of-sample test of H1**: fetch 2025–26 Breeze history (`analysis/history_fetch.py`; needs credentials) and confirm lot sizes, then run `oos_run.py` unchanged	Pass / fail of a pre-registered hypothesis on unseen data, with rank stability and reality-check results	The only path to genuine alpha evidence; everything so far is in-sample
2	**Obtain additional Algo 8 trade exports** (months before June or after October 2026) and re-apply the frozen findings unchanged (schedule, stop band, cycle stop, Clock-B timing)	A table of which reconstructed rules hold on new months	The only route to out-of-sample validation of the reconstruction
3	**Finish Phase 2 Track B screening** with the pre-registered register (five families; 21 primary tests; BH / Holm)	Reject / keep list with costs and stability	Quickly eliminates or promotes simple market hypotheses; inexpensive to run
4	**Finish Phase 2 Track A** (seven panels; family gains over baseline B2 with nulls)	A clean answer to "does any OHLCV family add information beyond the known premium mechanics?"	Closes the indicator question on clean data; the prior is negative
5	**Explain the post-cycle-stop short fills** (ALC-09) with existing data, and request order-lifecycle records	Whether they are in-flight orders or a behaviour of the stop module	Affects the cycle-stop rule, the most important risk control

12. Technical appendix

12.1 PRs, branches and commits

All PRs are open **drafts**; none is merged. Heads verified against GitHub on 11 November 2026.

PR	Branch	Base	Head	Content
#1	`research/ml-decision-engine`	`research/strong-findings` (`dee20138`)	`c5c4f285`	Lineage X ML decision engine
#2	`claude/friendly-galileo-r3zezw`	`main` (`5d6cc5cd`)	`07eee688`	Lineage Y decode, alpha anatomy, backtester, frozen H1
#3	`research/algo8-exit-engine`	PR #2	`6b4ef390`	Exit engine on corrected data
#4	`audit/algo8-reconstruction-audit`	PR #3	`28739d50`	Reconstruction audit and reconciliation
#5	`research/algo8-allocation-reconstruction`	PR #4	`4fab9d8f`	Allocation reconstruction phases 1–3
#6	`research/algo8-ohlcv-indicator-alpha`	PR #5	`aec2e090`	OHLCV foundation (Phases 0–1)

Lineage X's other branch `data/local-breeze-fetch` (`34c21f27`) holds the Breeze market data (639 MB), ~90 research scripts, the rulebook and Rahul's earlier report (`Rahul_Algo8_Reverse_Engineering_Report.pdf`); `research/strong-findings` (`dee20138`) holds `STRONG_FINDINGS.md` v7.

12.2 Key files

Area	Files
Raw input	`Trades_GKOHNP140.csv`, `Info.txt`
Lineage Y findings	`FINDINGS.md`; `analysis/*.py` (decode, indicators, clone, alpha search); `alpha_freeze/`
Exit engine	`analysis/exit_engine/EXIT_ENGINE_FINDINGS.md`; `results/exit_model_metrics.csv`, `first_passage_*.csv`, `autonomous_replay_metrics_{open,delay1,adverse}.csv`, `robustness_results.csv`, `order_key_audit.csv`
Audit	`audit/ALGO8_COMPLETE_RECONSTRUCTION_AUDIT.md`, `ALGO8_MASTER_FINDINGS_REGISTER.md`, `ALGO8_STRATEGY_SPECIFICATION.md`, `ALGO8_RECONSTRUCTION_READINESS.md`, `ALGO8_DATA_GAPS_AND_REQUIREMENTS.md`, `ALGO8_RECONSTRUCTION_ROADMAP.md`, `audit/evidence/`
Allocation	`analysis/allocation/ALLOCATION_FINDINGS.md`, `REPLACEMENT_REENTRY_FINDINGS.md`; `results/stop_ref_*.csv`, `replacement_*.csv`, `position_lifecycle_ledger.csv`, `state_transition_metrics.csv`
OHLCV	`ohlcv_research/data_manifest.md`, `OHLCV_FEATURE_CATALOG.md`, `OPPORTUNITY_PANEL_DESIGN.md`, `features.py`, `tests/test_features.py`, `results/`

Area	Files
This report	`report/ALGO8_COMPLETE_RESEARCH_REPORT.md` / `.pdf`, `report/ALGO8_CALL_BRIEF.md`, `report/assets/make_charts.py` (plots existing CSVs only)

12.3 Important statistics

Quantity	Value
Fills / orders / cycles	41,421 / 39,874 / 35
Short buy-back orders / lots	11,007 / 15,327
Short-opening sales (all modules)	13,410 orders, 15,312 lots
Contract-minute panel (PR #3)	205,752 rows (DEV 106,389 / VAL 43,679 / TEST 55,684)
PR #1 dataset	1,280,181 rows
Index bars / option bars	63,678 / 1,230,065
Split	DEV 18 cycles (≤ 20 Aug), VAL 8 (≤ 22 Sep), TEST 9

12.4 Definitions

Term	Definition
D-1 / D0	The pre-expiry session and the expiry session of a weekly cycle
BTST	Buy today, sell tomorrow; here, positions sold on D-1 and carried into D0
ATM / OTM	At-the-money (strike nearest the index) / out-of-the-money
Wing	A far-OTM long option bought as protection for a short
Lot / base unit	Exchange lot (NIFTY 65, SENSEX 20); Algo 8 sends one lot per child order
₹ L	Lakh = ₹100,000
MTM	Mark-to-market P&L at last traded prices
Recovered quote	limit ÷ 1.15 (buys) or ÷ 0.85 (sells): the price the algo saw when it decided
First passage	The first minute in which a price crosses a threshold
Exit association / stop-consistent	See §4 (measures 1 and 2)
Precision / recall	Share of predicted events that happened / share of actual events predicted
PR-AUC (AP)	Area under the precision-recall curve; compare it with the base rate, not with 0.5
DEV / VAL / TEST	Chronological whole-cycle splits: fit / threshold / score-once
Mode A / Mode B replay	Replay with actual entries and model exits / fully autonomous replay
Ownership convention	Which lot a buy-back is assumed to close: cheapest-first, FIFO or LIFO
Slot / chain allocation	Hypothesised allocation by schedule window / by recurring timer phase
BH-FDR	Benjamini–Hochberg false-discovery-rate control for many tests
Reality check / PBO	White's reality check for data snooping; probability of backtest overfitting
Readiness index	Subjective, unweighted mean of 19 component scores (§8)